

# Continuity

ANSWER KEY



## Testing continuity

- 1 Consider the piecewise function  $f(x) = x^2 + 1$  for  $x < 2$ , and  $f(x) = 3x - 1$  for  $x \geq 2$ .
    - a Find  $\lim_{x \rightarrow 2^-} f(x)$ . = 5.
    - b Find  $\lim_{x \rightarrow 2^+} f(x)$ . = 5.
    - c Is  $f$  continuous at  $x = 2$ ? Justify. Yes — both one-sided limits equal  $5 = f(2)$ , so  $f$  is continuous at  $x = 2$ .
  - 2 The piecewise function  $f(x) = kx + 1$  for  $x \leq 3$ , and  $f(x) = x^2 - 2$  for  $x > 3$ , is continuous at  $x = 3$ . Find the value of  $k$ .  
Need  $3k + 1 = 9 - 2 = 7$ , so  $k = 2$ .
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## Types of discontinuity and the Intermediate Value Theorem

- 3 Classify the discontinuity of each function at the given point:
    - a  $f(x) = \frac{x^2 - 1}{x - 1}$  at  $x = 1$  Removable (a hole) — the factor  $(x - 1)$  cancels.
    - b  $f(x) = \frac{1}{x - 2}$  at  $x = 2$  Infinite discontinuity —  $f$  blows up as  $x \rightarrow 2$ .
    - c  $f(x) = \lfloor x \rfloor$  at  $x = 3$  Jump discontinuity — the floor function jumps from 2 to 3.
  - 4 Show that  $f(x) = x^3 - 4x + 1$  has a root in the interval  $(1, 2)$ , using the Intermediate Value Theorem.  
 $f(1) = 1 - 4 + 1 = -2 < 0$  and  $f(2) = 8 - 8 + 1 = 1 > 0$ . Since  $f$  is continuous (a polynomial) and changes sign, the IVT guarantees a root in  $(1, 2)$ .
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## Continuity on an interval

- 5 State the largest interval(s) on which each function is continuous:
    - a  $f(x) = \frac{1}{x - 4}$  Continuous everywhere except  $x = 4$ : on  $(-\infty, 4) \cup (4, \infty)$ .
    - b  $f(x) = \sqrt{x - 3}$  Continuous on  $[3, \infty)$ .
    - c  $f(x) = \ln(x + 2)$  Continuous on  $(-2, \infty)$ .
  - 6 Find the value of  $c$  that makes  $f(x) = \frac{x^2 - 9}{x - 3}$  for  $x \neq 3$ , and  $f(3) = c$ , continuous at  $x = 3$ .  
 $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$ , so  $c = 6$ .
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## Applying the Intermediate Value Theorem

- 7 Show that  $f(x) = \cos x - x$  has a root in the interval  $(0, 1)$ .  
 $f(0) = \cos 0 - 0 = 1 > 0$  and  $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$ . Since  $f$  is continuous and changes sign on  $[0, 1]$ , IVT guarantees a root in  $(0, 1)$ .
- 8 A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.  
 Elevation as a function of time is continuous (the hiker moves without teleporting), starts below 500 m and ends above 500 m, so by the IVT the hiker's elevation must equal 500 m at some intermediate time.

## Visualizing discontinuities

- 9 The function  $f(x) = \frac{1}{x-2}$  (with an infinite discontinuity at  $x = 2$ ) is graphed below. Use the graph to describe the one-sided behavior near  $x = 2$ .  
 As  $x \rightarrow 2^-$ ,  $f(x) \rightarrow -\infty$ ; as  $x \rightarrow 2^+$ ,  $f(x) \rightarrow +\infty$  — the function is unbounded on both sides near  $x = 2$ , confirming an infinite (non-removable) discontinuity.

## Continuity of composite and combined functions

- 10 Explain why the composition of two continuous functions is continuous, and use this to argue that  $h(x) = \sqrt{x^2 + 1}$  is continuous everywhere.  
 If  $f$  is continuous at  $a$  and  $g$  is continuous at  $f(a)$ , then  $g \circ f$  is continuous at  $a$  (limits pass through continuous functions). Here  $x^2 + 1$  is a continuous polynomial (always  $\geq 1 > 0$ ), and  $\sqrt{u}$  is continuous for  $u \geq 0$ , so the composition  $\sqrt{x^2 + 1}$  is continuous for all real  $x$ .

## Removable discontinuities and redefinition

- 11 The function  $f(x) = \frac{x^2 + x - 6}{x - 2}$  has a removable discontinuity at  $x = 2$ . Find the value that, if assigned to  $f(2)$ , would make  $f$  continuous there.  
 Simplify:  $\frac{(x-2)(x+3)}{x-2} = x + 3$  for  $x \neq 2$ .  $\lim_{x \rightarrow 2} f(x) = 2 + 3 = 5$ . Assigning  $f(2) = 5$  removes the discontinuity.
- 12 Find all values of  $x$  where  $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$  has a discontinuity, and classify each as removable or non-removable.  
 Factor:  $\frac{(x-4)(x+4)}{(x-4)(x+2)}$ . Common factor  $(x-4)$ : removable discontinuity at  $x = 4$ . Remaining denominator zero at  $x = -2$ : non-removable (infinite) discontinuity.

## The Extreme Value Theorem (a continuity consequence)

- 13** State the Extreme Value Theorem, and explain why  $f(x) = x^2$  is guaranteed to have an absolute maximum and minimum on  $[-3, 2]$ .

The EVT states a continuous function on a closed interval  $[a, b]$  attains both an absolute max and min on that interval. Since  $f(x) = x^2$  is continuous (a polynomial) and  $[-3, 2]$  is closed, the EVT guarantees both extrema exist (here: min at  $x = 0$ , max at  $x = -3$ ).